DIY Spectrometer

Materials:

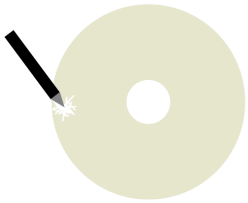
- 1 CD, DVD, or diffraction grating
- 1 toilet paper roll
- Cardboard or paper (to fill the ends of the toilet paper roll)
- Masking Tape & Packing Tape
- Scissors & X-Acto Knife/box cutter
- Pencil

INSTRUCTIONS

Step 1: Preparing the CD/DVD/Diffraction Grating

Option A: CD Preparation (easier)

- Use an X-Acto Knife to scrape off a tiny section of the thin cover near the edge of the CD.

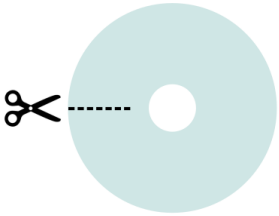


- Take a piece of packing tape and stick it onto the section that you just scraped off. Then peel the tape off and the cover should come off with the tape.

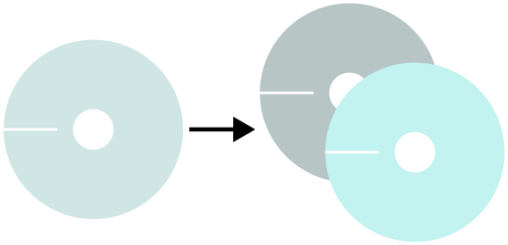
- Repeat this for the entire CD until the cover is fully removed. What now remains is the part that will be used as the diffraction grating for the spectroscope!

Option B: DVD Preparation (harder)

- Take a pair of scissors and make a cut towards the center of the DVD. The cut does not have to go all the way through (see picture below)

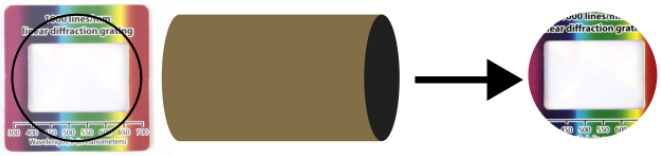


- Then from the cut that you just made, carefully split the DVD into its two layers with your fingernails. The layer that we want is the clear layer (without the top sticker) as the diffraction grating!



Option C: Diffraction grating preparation

- Using a pencil, trace a circle around the diffraction grating with the toilet paper roll. Using scissors, cut around the circle you just traced.

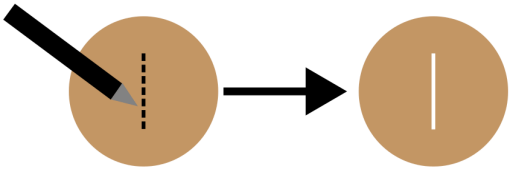


Step 2: Preparing the ends of the spectroscope

a) Place one end of the toilet paper roll on your cardboard or paper and trace out 2 circles. Cut both circles out with your scissors.



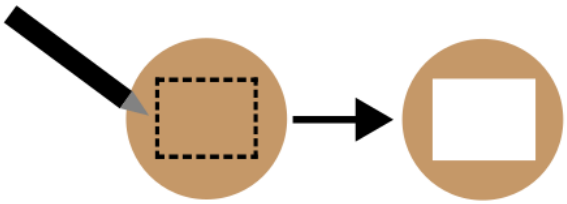
b) Circle #1 will be the slit end. Take your X-Acto knife or box cutter and cut a thin slit at the middle of the circle about half of the diameter of the circle.



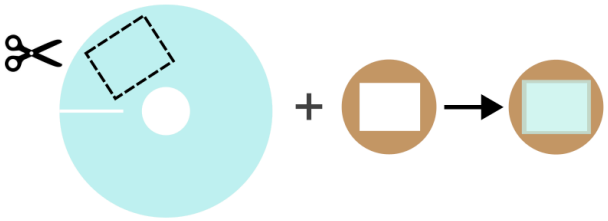
c) Circle #2 will be the diffraction grating end.

   i) If you're using a CD/DVD:

      1) Draw a rectangle inside the circle (the corners should not reach the edges!) with a pencil wide enough for you to view through and cut out the rectangle from an X-Acto knife or box cutter to create a window in the circle.



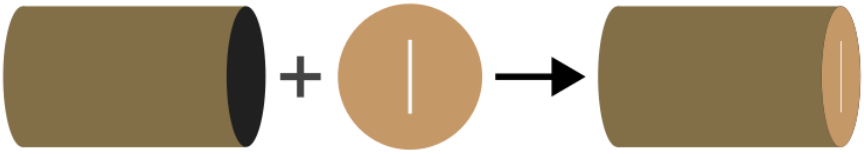
2) Then cut a section of the CD/DVD big enough to cover the window and tape it to the window. Use masking tape to cover up the sharp edges.



ii) If you're using a pre-bought diffraction grating, you can simple use the cutout from Step 1 Option C (or use cardboard/paper to cover up the edges if you want)

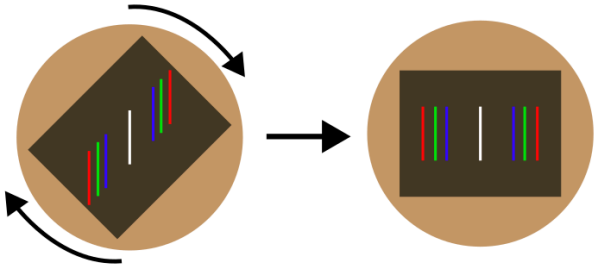
Step 3: Assembling the spectroscope

a) Tape the circle with the slit to one end of the toilet paper roll with masking tape.

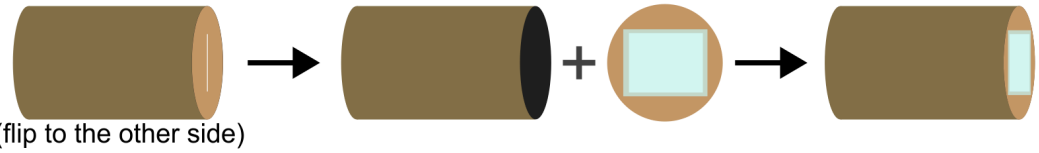


b) Before taping the diffraction grating to the other end of the toilet paper roll, hold up the spectroscope and diffraction grating so that you're viewing the diffraction grating side and the slit is on the other side. Point it towards a light source **(not the sun!)** and you should see some rainbow lines!

c) Rotate the diffraction grating so that the lines are clearly split across from the slit (see picture below).



d) Once you find the right orientation, tape it to the toilet paper roll.



(flip to the other side)